Pratham Aggarwal

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Education

University of California San Diego Expected Mar 2027
Bachelor of Science in Mathematics-Computer Science; Data Science
Honors: HDSI Scholarship Recipient, HKN-IEEE Award

Skills

Programming Languages Python, SQL, C, C++, Java, JavaScript, TypeScript
AI / Machine Learning PyTorch, TensorFlow, Scikit-Learn, PowerBI, Tableau, Matplotlib, Pandas, Computer Vision
Systems & Tools Linux, Docker, Kubernetes, Google Cloud Platform (GCP), Git/GitHub, Shell Scripting

Experience

Machine Learning Researcher @ Belkin Lab, UC San Diego Apr 2026 – Present
- Investigating **controllable generation** in masked diffusion language models (MDLMs) via inference-time unmasking, developing **masking-aware activation steering** methods for partially observed sequences.
- Benchmarking **token selection strategies** (margin, entropy, saliency, concept-directed) on controllability and text quality using perplexity, MAUVE, and classifier-based metrics on the LLaDA codebase.

Machine Learning Researcher @ Scripps Institution of Oceanography Jun 2025 – Present
- Built a neural climate modeling pipeline achieving **$\sim$20$\times$ speedup** over state-of-the-art by coupling a lightweight UNet with an ocean emulator, maintaining prediction accuracy over a **40-year** autoregressive rollout.
- Trained a conditional diffusion model (**DDPM**) on **155 years of simulation data** to generate **20–50 member ensembles** per timestep, recovering distributional extremes and fine-grained spatial structure lost in standard regression baselines.
- Designed an ablation study across **4 atmospheric configurations** to identify the model with minimum complexity.
- Deployed a full-stack ML evaluation platform on **Google Cloud** benchmarking **500+ model outputs** with **20+ custom metrics**, earning a **$4,500 scholarship** for advancing model interpretability tooling.

Full Stack Software Developer Intern @ NIOV Labs Corporation Jan 2026 – Apr 2026
- Built an NLP-based system to extract goals, tasks, participants, and deadlines from meeting transcripts, achieving **88%** agreement with human-labeled results across **20+** validation transcripts.
- Evaluated automated scheduling using extracted meeting details and calendar availability, cutting manual scheduling effort by **60%** while achieving 95% successful confirmations with 2-second response times.
- Containerized the application with **Docker** and deployed it on **Kubernetes**, improving reliability and enabling scalable deployments across environments.

Robotics Research Assistant @ Existential Robotics Lab Aug 2025 – Dec 2025
- Benchmarked and optimized motion-planning algorithms (A*, RRT, MPC), achieving a **28%** reduction in path-planning failures and maintaining average trajectory error below **0.15m** in dynamic environments.
- Enhanced PyBullet-based simulation pipelines for mapping and localization, validating performance across **100+** experimental trials.
- Improved simulation speed by **2.1x** using physics evaluations, enabling concurrent testing of **10+** planning algorithms.

Quantitative Researcher @ SFIC Quantitative Technologies Mar 2025 – Jun 2025
- Improved simulated options trading returns by **12%** for **$1.3M** student-run investment fund by developing a Deep Learning agent with adaptive policy learning and early exercise strategies.
- Improved trading decisions by **18%** through a sequential decision-making step that dynamically selected option type, expiration, and exercise timing from historical time-series data.
- Reduced stock price prediction error to **0.96 MSE** by training an **LSTM** model on **5 years** of market data with hyperparameter tuning and feature engineering.

Data Science Intern @ Solana Center Jan 2025 – Mar 2025
- Led a **5-member** data analytics team to process **2,000+** composting records using **Python** and **SQL**, developing a real-time **Power BI** dashboard that improved evaluation accuracy by **6.7%** and guided program decisions.
- Boosted dataset reliability by **9%** through automated data manipulation, cleaning and imputation routines, ensuring consistent and reproducible analytics across stakeholders.
- Managed team workflows and version control using **Linux** and **Git**, improving collaboration and reporting efficiency.

Projects

NSF Hackathon (1st Place): Coastal Flooding Prediction | ***CatBoost, Feature Engineering*** Jan 2026
- Built and evaluated ML models to predict coastal flooding risk during a CERN-hosted hackathon, selecting CatBoost as the best-performing approach and achieving a **0.94 F1 score** on the final benchmark.
- Improved flood prediction accuracy by comparing multiple models and tuning features on large-scale data, validating results through competition-based evaluation on CodaBench.

Hackfrontier (1st Place): Homeless Services Computer Vision Platform | ***Scikit-learn, Pandas, OpenCV*** Jun 2025
- Built a geospatial forecasting platform that integrated **35+** transit, demographic, and geographic features, achieving **67%** accuracy in optimizing placement of homeless service centers.
- Deployed a real-time computer vision system with Oxen.ai and EyePop.ai to track transient populations, delivering actionable insights for data-driven service allocation.